# RE-PREDICT

## AI-Powered Renewable Energy Intelligence and Forecasting Platform

Complete Product, Technical and Hackathon Master Document

*Sense. Understand. Predict. Detect. Simulate. Explain.*

Prepared for Zinnovatio 4.0 - Energy, Environment and Climate Action Track

Purpose: judge-ready concept brief, developer blueprint and source document for the final presentation

Status legend: CURRENT MVP = demonstrable now; PROPOSED = designed for build; FUTURE = post-MVP extension

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# 1. Executive Summary and Identity

## 1.1 Executive summary

RE-PREDICT is a proposed renewable-energy intelligence platform that turns officially published renewable-generation reporting into transparent, decision-support outputs. It is designed to ingest approved source reports, preserve provenance, normalize time-series data, visualize trends, forecast future generation, flag unusual observations, simulate capacity scenarios and communicate assumptions in plain language. The product does not replace official reporting or operational dispatch decisions. It is an independent analytical layer built on top of source data and clearly separates observed values, model estimates and user-entered scenarios.

For Zinnovatio 4.0, the current MVP should prove a narrow but credible flow: source report to validated dataset to dashboard to baseline forecast/anomaly view to documented scenario. All performance, impact and emissions figures remain unreported until measured using frozen datasets and declared methodology.

## 1.2 Project identity

| Element | Definition |
|---|---|
| Name | RE-PREDICT |
| Tagline | Sense. Understand. Predict. Detect. Simulate. Explain. |
| Positioning | Independent renewable-energy intelligence and decision-support layer. |
| Primary proposition | Make reported renewable-generation data traceable, understandable and actionable without misrepresenting it as official CEA output. |
| Product boundary | Analytics and planning support; not a replacement for CEA, grid operators, weather services or investment advice. |

## 1.3 Elevator pitches

One line: RE-PREDICT transforms official renewable-generation reports into traceable forecasts, anomaly signals and scenario insights for clearer energy decisions.

Three lines: Renewable-energy reporting explains what has already happened, while planning also needs an evidence-led view of what may happen next. RE-PREDICT preserves source provenance, analyzes historical patterns and exposes forecasts, anomalies and what-if scenarios with explicit uncertainty. It is a transparent decision-support layer, not an official reporting system or control room.

# 2. Track Alignment and Problem Landscape

## 2.1 Zinnovatio 4.0 alignment

The concept aligns with the Energy, Environment and Climate Action track by addressing renewable-energy visibility, variability and planning support. Its value proposition is not a claimed impact result: it is a proposed capability to help users interpret renewable-generation patterns and explore planning assumptions with auditable evidence.

## 2.2 Problem statement

Renewable generation varies across time because underlying conditions, installed capacity, operational events and reporting granularity vary. Historical reports are essential, but they do not by themselves provide a forecast, quantified uncertainty, systematic anomaly signal, scenario comparison or consistently explained analytical narrative. Users therefore need a trusted bridge from reporting to analysis while retaining source context and limits.

| Challenge | Consequences | RE-PREDICT response |
|---|---|---|
| Historical reporting | Users must manually interpret trends and changes. | Trend views and source-linked analytics. |
| Uncertain future output | Planning conversations lack a common forecast baseline. | Time-aware forecasts with intervals and error metrics. |
| Unusual values are hard to triage | A change may be data quality, capacity, weather or operations. | Anomaly flags plus cautious evidence-based explanation. |
| Scenario questions are fragmented | Capacity assumptions are often compared informally. | Explicit what-if simulator with assumptions. |
| Climate claims can be overstated | Carbon estimates depend on a defensible factor and boundary. | Methodology-first estimates; no unsupported impact claims. |

# 3. CEA Data Foundation and Positioning

## 3.1 Verified source positioning

The Central Electricity Authority (CEA) is identified on the Renewable Generation Reports portal as the Central Electricity Authority under the Ministry of Power, Government of India. The portal labels a “Daily Renewable Generation Report” and provides report names and report dates for download. These are the only CEA-specific claims made in this document; no current totals, fields, methodology, coverage or performance claims are assumed without inspecting the relevant report and its metadata.

CEA reporting is the authoritative published source in this product design because it is the original government publisher of the cited portal material. RE-PREDICT must never relabel its derived values as CEA values, imply endorsement, or overwrite source files. Every displayed derived output should expose its source report identifier, download time, transformation version and data-quality status.

## 3.2 Data contract and provenance

```
Official CEA report -> immutable raw file -> extraction run -> normalized records
     -> validation results -> curated time series -> analytics/model outputs
     -> API/dashboard with source and assumption labels
```

| Layer | Required controls |
|---|---|
| Raw | File checksum, source URL, report name/date, retrieval timestamp, parser version, immutable storage. |
| Extracted | Captured cells/tables, extraction confidence, page/sheet locator where possible. |
| Normalized | Canonical units/timezone, technology/location labels, source-to-field mapping. |
| Curated | Completeness, duplicates, range checks, reconciliation notes and approval status. |
| Derived | Model version, training cutoff, feature version, metric snapshot and scenario assumptions. |

# 4. Solution and Users

## 4.1 Sense to Explain framework

| Stage | Meaning | Output |
|---|---|---|
| Sense | Acquire source reports and preserve provenance. | Raw files, metadata, validation status. |
| Understand | Normalize and visualize historical observations. | Trends, comparisons, quality indicators. |
| Predict | Estimate a defined future target from prior information. | Point forecast and interval. |
| Detect | Identify observations that depart from expected patterns. | Flag, score and evidence. |
| Simulate | Test declared assumptions such as capacity scaling. | Scenario comparison, not a prediction. |
| Explain | Translate evidence, assumptions and uncertainty for users. | Cited narrative with guardrails. |

## 4.2 Target users and use cases

| User | Need | MVP use case |
|---|---|---|
| Policy and planning teams | Compare patterns and scenario assumptions. | Review trend, forecast and capacity-change scenario. |
| Utilities and analysts | Triage changes against an expected range. | Investigate source-linked anomaly flag. |
| Researchers and students | Reproducible access to transformed data. | Download provenance-rich curated dataset. |
| Sustainability teams | Understand methodology for proxy carbon estimates. | Inspect factor, boundary and caveats. |
| Hackathon judges and mentors | Assess technical credibility and clarity. | Walk end-to-end source-to-insight demo. |

# 5. Product Modules and User Experience

## 5.1 Complete product modules

| Module | CURRENT MVP | PROPOSED / FUTURE |
|---|---|---|
| Source Studio | Source register and upload/download audit trail. | Automated scheduled retrieval with parser monitoring. |
| Data Quality | Basic schema, missingness, duplicate and range checks. | Cross-report reconciliation and human review workflow. |
| Insights Dashboard | Trend charts, filters, source labels. | Comparative regional/technology analysis when verified fields permit. |
| Forecast Lab | Baseline and tree-model evaluation on selected target. | Operational retraining, intervals and model selection. |
| Anomaly Desk | Residual/z-score flags and status. | Multivariate detector with analyst feedback loop. |
| Scenario Studio | Capacity-scaling calculation with assumptions. | Constraint-aware multi-variable scenarios. |
| Carbon Lens | Methodology shell and disclosed factor input. | Validated location/time-specific factors if licensed and governed. |
| AI Energy Analyst | Grounded summaries of selected outputs. | Conversational planning assistant with expanded retrieval. |

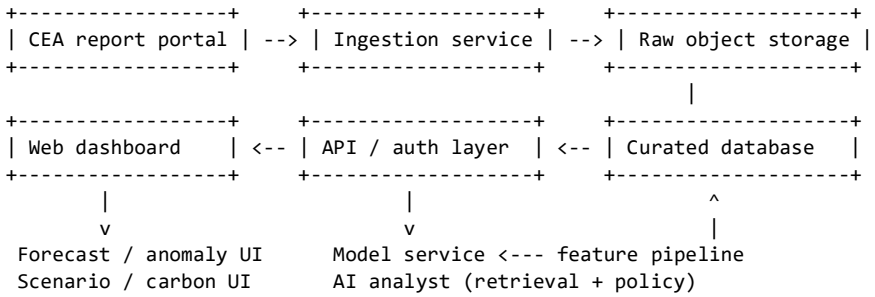
## 5.2 Page-by-page dashboard specification

| Page | Core elements | Trust UX |
|---|---|---|
| Home | Purpose, last refresh, dataset status, quick links. | Status legend and source disclaimer. |
| Data Sources | Report inventory, source links, checksums, parser status. | Open source file and transformation lineage. |

| | | |
|---|---|---|
| Insights | Date/technology/location filters, trend and comparison charts. | Observed vs derived visual distinction. |
| Forecast | Horizon, target, actual-vs-predicted, interval, metrics. | Training cutoff, validation window and model card. |
| Anomalies | Flag list, expected band, residual, review state. | “Signal, not cause” language and evidence links. |
| Scenarios | Inputs, equations, side-by-side cases. | Assumptions highlighted; no forecast label. |
| Carbon Lens | Generation basis, factor, boundary, calculation. | Estimate label and caveats. |
| AI Analyst | Question input, cited answer, evidence panel. | Refuse unsupported claims; show sources. |
| Admin | Ingestion jobs, quality exceptions, model registry. | Role-based access and audit logs. |

# 6. Architecture and Data Foundation

## 6.1 End-to-end architecture




## 6.2 Ingestion, extraction, normalization and validation

The ingestion worker registers the official URL and report metadata, retrieves the file only under approved access rules, calculates a checksum and stores an immutable copy. A format-specific extractor produces a structured staging record. A normalization layer maps only observed verified fields into a versioned canonical schema; unknown fields remain unmapped. Validation blocks promotion when required keys, dates, units or ranges fail policy. Human review is required for low-confidence extraction or changed layouts.

## 6.3 Proposed database schema

| Entity | Key fields | Purpose |
|---|---|---|
| source_reports | report_id, source_url, source_name, report_date, retrieved_at, checksum | Immutable source registry. |
| ingestion_runs | run_id, report_id, parser_version, status, log_uri | Trace extraction and retries. |
| generation_observations | observation_id, timestamp, technology, geography, value, unit, report_id | Normalized observation fact table. |
| quality_results | result_id, observation_id/run_id, rule_id, severity, status | Quality evidence. |
| feature_sets | feature_set_id, cutoff, definition_version | Reproducible model inputs. |
| model_runs | model_run_id, algorithm, training_range, metrics, artifact_uri | Model registry. |
| forecasts | forecast_id, model_run_id, target_time, point, lower, upper | Forecast results. |
| anomaly_events | event_id, observation_id, score, method, review_status | Anomaly signal and review. |
| scenarios | scenario_id, inputs_json, equation_version, created_by | Scenario assumptions and results. |
| audit_events | event_id, actor, action, object_ref, timestamp | Governance audit trail. |

# 7. Analytics, Forecasting and Validation

## 7.1 Forecast problem formulation

For a chosen geography, technology and aggregation level, define the target y(t+h) as generation at horizon h using only information available at or before cutoff t. The MVP must select one specific target and time granularity supported by validated data. It must not train on future fields, revised values not available at cutoff, or synthetic labels presented as observed data.

```
y_hat(t+h) = f( lagged generation, calendar features, verified exogenous features, capacity proxy if
available )
```

## 7.2 Feature engineering

| Feature family | Examples | Safeguard |
|---|---|---|
| Autoregressive | y(t-1), y(t-7), rolling mean/median. | Compute using past data only. |
| Calendar | hour/day/week/month, holiday flag where approved. | Encode cyclic variables if useful. |
| Seasonality | Fourier terms or seasonal indicators. | Fit only on training history. |
| Capacity/context | Verified installed-capacity or source flags if present. | Version field and record availability date. |
| Weather | Only licensed/validated external data. | FUTURE unless integrated and provenance-complete. |

## 7.3 Models and validation

Baseline models are mandatory: seasonal naive, persistence and moving average. Proposed supervised candidates are XGBoost and LightGBM regression trained on lagged/calendar features. They are compared against baselines, not assumed superior. LSTM and Temporal Fusion Transformer are FUTURE options only after data volume, feature governance, compute cost and business value justify them.

```
Walk-forward validation:
train [t0..t1] -> test [t1+1..t2]
train [t0..t2] -> test [t2+1..t3]
... aggregate metrics across ordered folds
```

| Metric | Equation / interpretation |
|---|---|
| MAE | mean(\|y - y_hat\|); average absolute error in target units. |
| RMSE | sqrt(mean((y - y_hat)^2)); weights large errors more strongly. |
| sMAPE | mean(2\|y-y_hat\|/(\|y\|+\|y_hat\|+epsilon))*100; define epsilon. |
| R2 | 1 - sum((y-y_hat)^2)/sum((y-mean(y))^2); interpret cautiously for time series. |
| Interval coverage | fraction of y within [lower, upper]; compare to nominal coverage. |
| Interval width | mean(upper-lower); report alongside coverage, not alone. |

Acceptance of a model requires predeclared folds, a frozen dataset version, baseline comparison, no leakage check and metrics reported by horizon. No score is claimed in this document.

# 8. Anomalies, Scenarios and Carbon Intelligence

## 8.1 Anomaly detection design

An anomaly is an observation that deviates from a defined expected pattern; it is not proof of a physical cause. The MVP may use residual scoring against a seasonal baseline or forecast. For observed y(t), expected y_hat(t) and robust residual scale s, define r(t)=y(t)-y_hat(t) and score a(t)=|r(t)|/(s+epsilon). Flag when a(t) exceeds a calibrated threshold k, with k chosen from validation history and documented.

```
r(t) = y(t) - y_hat(t)
a(t) = |r(t)| / (s + epsilon)
flag(t) = 1 if a(t) > k, else 0
```

Explanation constraints: the system may say “generation was below the model’s expected range given the available inputs.” It must not attribute cause to weather, outage, curtailment, policy or equipment failure unless that evidence is supplied by an approved source. The UI should list candidate factors as hypotheses, distinguish source facts from model inference and route high-severity flags to human review.

### 8.2 Scenario simulator

A simple capacity-scaling scenario is a transparent planning calculation, not a forecast. If baseline generation G is associated with a declared baseline capacity C0 and a scenario capacity C1, then scenario output is G_s = G x (C1/C0) x A, where A is an explicit availability/derate assumption. If capacity is not source-verified, the scenario must remain unavailable or use a clearly user-entered proxy.

```
G_s = G x (C1 / C0) x A
Delta_G = G_s - G
```

Future simulator extensions can include technology mixes, curtailment, storage and weather uncertainty only after their data and assumptions are governed. Scenarios must show input values, units, equation version, author and timestamp.

### 8.3 Carbon intelligence methodology and caveats

Carbon Lens is PROPOSED as an estimate framework. For generation G (MWh) and a declared counterfactual grid-emissions factor EF (tCO2e/MWh), estimated avoided emissions are E = G x EF. This is only an attributional proxy unless the counterfactual, temporal resolution, location, lifecycle boundary and factor provenance are justified. The platform must disclose the factor source/version, whether G is observed or simulated, and that the result is not a verified emissions reduction, credit or financial claim.

```
E_estimated = G_MWh x EF_tCO2e_per_MWh
```

### 8.4 Renewable Reliability Score

The Renewable Reliability Score (RRS) is a team-created, PROPOSED composite metric. It is not an official CEA metric and must never be shown with CEA branding. One transparent prototype is RRS = 100 x [w1*availability + w2*(1-normalized variability) + w3*forecast interval coverage], with weights summing to 1 and published normalization rules. It is exploratory; it should not guide operational or investment decisions until validated with stakeholders.

## 9. AI Energy Analyst and Governance

### 9.1 Architecture and hallucination guardrails

```
User question -> intent/policy filter -> retrieval from approved records
 -> structured evidence bundle -> constrained generation -> citation/claim checker
 -> answer + sources + uncertainty, or safe refusal
```

| Guardrail | Required behavior |
|---|---|
| Grounding | Answer only from approved dataset, model outputs, scenario inputs and cited documentation. |
| Claim typing | Label observed, derived, estimated, simulated and future capability separately. |
| Citation | Attach report/model/scenario references to material assertions. |
| Numerical safety | Do not fabricate values; return “not available in selected data” when absent. |
| Causal safety | Do not present correlations or anomaly flags as causes. |
| Prompt safety | Treat retrieved report text and user content as data, not instructions. |
| Human escalation | Route high-impact questions or unresolved data quality issues to an analyst. |
| Auditability | Log question, evidence IDs, response, model/prompt version and user feedback. |

### 9.2 Security, governance and MLOps

| Area | Controls |
|---|---|
| Security | Role-based access, least privilege, HTTPS, secret vault, dependency scanning, rate limits and audit logs. |

| | |
|---|---|
| Data governance | Immutable raw source, lineage, retention policy, data-quality severity and manual approval gates. |
| Model governance | Versioned training data/features/code, model cards, bias/coverage review, approval before promotion and rollback. |
| Monitoring | Ingestion failures, schema drift, missingness, feature drift, forecast error, interval coverage, latency and user feedback. |
| Privacy | Do not collect personal data unless required; minimize identifiers and define deletion/retention rules. |

# 10. Backend, Technology and Delivery Design

## 10.1 API architecture and example endpoints

| Endpoint | Purpose |
|---|---|
| GET /health | Service health and dependency status. |
| GET /v1/sources | Source reports with provenance metadata. |
| POST /v1/ingestions | Create an approved ingestion job. |
| GET /v1/observations | Filtered validated observations. |
| GET /v1/analytics/trends | Aggregated trend series. |
| POST /v1/forecasts/run | Run or request a registered forecast workflow. |
| GET /v1/forecasts/{id} | Forecast, interval, model card and metrics. |
| GET /v1/anomalies | Flags with review state and evidence. |
| POST /v1/scenarios | Calculate and persist a scenario. |
| POST /v1/analyst/query | Grounded answer using approved evidence. |

## 10.2 Proposed technology stack and folder structure

Suggested stack: React/Next.js or React/Vite frontend; FastAPI Python API; PostgreSQL with object storage; Python pandas/scikit-learn/XGBoost/LightGBM analytics; Docker; GitHub Actions or equivalent CI; optional MLflow or a simple versioned model registry. Selection remains PROPOSED and should be adjusted to team capability and hackathon constraints.

```
re-predict/
  apps/web/                # dashboard
  services/api/            # FastAPI routes, auth, schemas
  services/worker/         # ingestion and scheduled jobs
  packages/data_contract/  # schemas, mappings, validation rules
  packages/analytics/      # features, models, evaluation
  infra/                   # Docker, deployment manifests
  tests/                   # unit, integration, e2e, acceptance
  docs/                    # model cards, data dictionary, runbooks
```

## 10.3 Testing strategy

Use unit tests for parsers, calculations, validation rules and API schemas; integration tests for raw-to-curated ingestion and persistence; contract tests for APIs; end-to-end browser tests for key journeys; reproducibility tests for model runs; security checks for authentication and secrets; and data-quality regression tests against frozen fixtures. Tests must use synthetic/sanitized fixtures where redistribution of source files is not allowed.

# 11. Build Plan, Workstreams and Priorities

## 11.1 Explicit build phases 0 to 16 and dependencies

| Phase | Name | Deliverable | Depends on |
|---|---|---|---|
| 0 | Foundation | Repo, environments, frontend/backend skeleton, /health. | None |

| | | | |
|---|---|---|---|
| 1 | CEA acquisition | Source registry, approved retrieval/manual intake, raw storage. | 0 |
| 2 | Extraction | Format parser and staging schema. | 1 |
| 3 | Normalization | Canonical data contract, unit/date mapping. | 2 |
| 4 | Validation and provenance | Quality rules, checksums, lineage UI/API. | 3 |
| 5 | Database and API | Curated storage and read APIs. | 4 |
| 6 | Insights dashboard | Filters, trend charts, source labels. | 5 |
| 7 | Baseline forecasting | Target definition, seasonal naive/persistence, walk-forward evaluation. | 5 |
| 8 | Tree-model forecasting | XGBoost/LightGBM comparison, model card. | 7 |
| 9 | Anomaly detection | Residual score, review workflow, evidence UI. | 7 |
| 10 | Scenario simulator | Capacity-scaling calculator and saved assumptions. | 5 |
| 11 | Carbon Lens | Methodology disclosure and controlled factor input. | 10 |
| 12 | AI Energy Analyst | Retrieval, policy filter, citations and refusal path. | 5, 8, 10 |
| 13 | Security and MLOps | Auth, audit, CI, registry, monitoring. | 5 onward |
| 14 | Integration and UX hardening | E2E flows, empty/error states, accessibility. | 6-13 |
| 15 | Demo and PPT | Script, fixtures, screenshots, judge material. | 14 |
| 16 | Release gate | Acceptance review, rollback plan, deployment rehearsal. | 15 |

## 11.2 Parallel team workstreams

| Workstream | Primary ownership | Can start |
|---|---|---|
| Data | Source registry, parser fixtures, contract, validation. | Phase 0; Phase 1 after foundation. |
| Backend | API schemas, database, job orchestration. | Phase 0/5 interfaces in parallel. |
| ML | Target definition, baselines, folds, anomaly prototype. | Phase 5 fixtures; use synthetic fixture early. |
| Frontend/UX | Dashboard shell, design system, empty/loading states. | Phase 0 with API mocks. |
| Product/demo | Story, slide content, model/data cards, judge Q&A. | Immediately; synchronize at each phase gate. |

## 11.3 P0 to P3 priority framework

| Priority | Scope |
|---|---|
| P0 - must demo | Source provenance, validated sample, insights page, baseline forecast, anomaly signal, scenario, disclaimer. |
| P1 - strengthen | Tree-model comparison, model card, AI Analyst grounded answer, e2e tests. |
| P2 - polish | Carbon Lens methodology, role-based admin, richer drill-downs, monitoring. |
| P3 - future | External weather integration, LSTM/TFT, constraint-aware simulator, production scale-out. |

## 11.4 48-hour hackathon plan

| Time | Outcome |
|---|---|
| 0-4h | Freeze scope, source contract, UI wireframe, fixtures, repository and owners. |
| 4-12h | Ingestion/normalization path, API skeleton, dashboard shell. |
| 12-22h | Validated dataset, insights page, baseline forecast and evaluation. |
| 22-30h | Anomaly/scenario modules, provenance and error states. |
| 30-38h | Integrate demo flow, model/data cards, grounded analyst prototype if P0 complete. |

| | |
|---|---|
| 38-44h | Acceptance tests, accessibility/performance fixes, rehearsal. |
| 44-48h | Final demo recording/rehearsal, PPT, release gate and contingency packaging. |

# 12. Concrete Acceptance Tests and Release Gate

Global rule: a phase is not complete until every listed mandatory test passes, failures are fixed and the test is rerun. Test records must state fixture/version, timestamp, owner and PASS/FAIL. “Built” is not proof of acceptance.

| Phase | Test ID and subject | PASS condition |
|---|---|---|
| 0 | AT-0.1 Frontend starts | Application page loads without compilation error. |
| 0 | AT-0.2 Backend starts | API starts without import/runtime error. |
| 0 | AT-0.3 Health endpoint | GET /health returns HTTP 200 and status ok. |
| 0 | AT-0.4 Separation | No frontend code directly accesses database; path is UI -> API -> service -> DB. |
| 1 | AT-1.1 Source registration | Official report URL, name/date, retrieval time and checksum are stored. |
| 1 | AT-1.2 Raw preservation | Retrieved fixture/file is immutable and retrievable by report ID. |
| 1 | AT-1.3 Failure safety | Unavailable source creates a failed job with retry/log; no fabricated data. |
| 2 | AT-2.1 Extract known fixture | Parser creates staging records matching expected fixture values. |
| 2 | AT-2.2 Layout change | Changed/unrecognized layout is quarantined with status, not silently parsed. |
| 3 | AT-3.1 Canonical mapping | Date, unit and category mappings match versioned contract. |
| 3 | AT-3.2 Unknown field | Unmapped field remains flagged, not guessed. |
| 4 | AT-4.1 Required checks | Missing key/date/unit fails promotion with visible rule ID. |
| 4 | AT-4.2 Duplicate check | Duplicate observation is detected or deterministically resolved with audit. |
| 4 | AT-4.3 Provenance trace | Curated record links to report ID, run ID and mapping version. |
| 5 | AT-5.1 Persistence | Validated fixture persists and query returns expected records. |
| 5 | AT-5.2 API contract | Invalid query returns documented 4xx response; valid query schema passes. |
| 6 | AT-6.1 Trend parity | Dashboard aggregate equals API aggregate for fixed fixture. |
| 6 | AT-6.2 Trust labels | Observed and derived values plus source link/status are visible. |
| 7 | AT-7.1 Leakage test | Feature build for cutoff t contains no records later than t. |
| 7 | AT-7.2 Baselines | Seasonal naive/persistence complete all ordered folds and metrics save. |
| 7 | AT-7.3 Reproducibility | Frozen fixture + config reproduces baseline output within tolerance. |
| 8 | AT-8.1 Fair comparison | Tree models use same folds/horizon and compare to baseline metrics. |
| 8 | AT-8.2 Model card | Training range, features, cutoff, metrics and limits are displayed. |
| 9 | AT-9.1 Known signal | Injected fixture anomaly triggers at calibrated threshold. |
| 9 | AT-9.2 No causal claim | UI labels flag as signal and does not assert unsupported cause. |
| 10 | AT-10.1 Equation parity | Scenario API equals G x (C1/C0) x A for test inputs. |
| 10 | AT-10.2 Assumption trace | Saved scenario exposes inputs, units, equation version and author/time. |
| 11 | AT-11.1 Carbon parity | Estimate equals G x EF with declared units. |
| 11 | AT-11.2 Caveat display | Factor source/version and non-verified-estimate caveat are visible. |
| 12 | AT-12.1 Grounded answer | Answer cites selected approved evidence IDs. |
| 12 | AT-12.2 Unsupported ask | No-evidence question returns safe limitation/refusal, not invented number. |
| 13 | AT-13.1 Access control | Unauthorized protected request returns 401/403; authorized path succeeds. |
| 13 | AT-13.2 Audit | Ingestion/model/scenario actions create immutable audit event. |

| | | |
|---|---|---|
| 14 | AT-14.1 E2E core flow | Source -> insight -> forecast -> anomaly -> scenario completes on fixture. |
| 14 | AT-14.2 Error states | Offline/empty/failed ingestion states are clear and do not crash. |
| 15 | AT-15.1 Demo rehearsal | Team completes scripted demo within allotted time on clean environment. |
| 15 | AT-15.2 Evidence pack | PPT claims link to source, model card or are marked proposed. |
| 16 | AT-16.1 Release checklist | All P0 tests pass; no unresolved critical/high issue. |
| 16 | AT-16.2 Rollback | Prior stable build or static evidence demo can be restored and demonstrated. |

### 12.1 Failure-safe rules and final release gate

| Rule | Required action |
|---|---|
| Source unavailable or parser uncertain | Show source/ingestion failure; quarantine output; do not substitute values. |
| Validation failure | Do not promote to curated/model/dashboard layer. |
| Model underperforms baseline | Keep baseline or label experiment; do not claim improvement. |
| Interval/quality failure | Hide or mark forecast unavailable until investigated. |
| Anomaly uncertainty | Show signal and evidence only; require human review for interpretation. |
| AI lacks evidence | Refuse or ask to narrow selection; never fabricate. |
| Security failure | Block release and rotate/revoke exposed credentials. |
| Final release gate | All P0 acceptance tests PASS; data/model cards complete; demo rehearsal passes; no critical/high unresolved defect; rollback and disclaimer verified. |

## 13. MVP, Roadmap, Deployment and Adoption

### 13.1 Scope distinction

| CURRENT MVP | PROPOSED next build | FUTURE roadmap |
|---|---|---|
| Validated limited-scope dataset, provenance view, trends, baseline forecast, residual anomaly, capacity scenario and transparent disclaimers. | Tree-model comparison, model cards, AI Analyst with grounding, governance automation and controlled carbon-methodology interface. | External validated weather/context feeds, LSTM/TFT where justified, broader coverage, operational monitoring, stakeholder validation and enterprise integrations. |

### 13.2 Deployment and commercialization/adoption possibilities

Deployment options are PROPOSED: a containerized web/API/worker stack in a controlled cloud or institutional environment; scheduled jobs with a manual fallback; managed database/object storage; and a static demo deployment for hackathon resilience. Adoption pathways may include research labs, educational institutions, public-interest data partnerships, utility analytics teams and sustainability teams. Any paid, public-sector or procurement model requires separate legal, data-rights, security and stakeholder validation; no revenue or savings figure is assumed.

## 14. Risks, KPIs and Judge Differentiation

### 14.1 Risks and limitations

| Risk / limit | Mitigation |
|---|---|
| Source schema or availability changes | Version parsers, monitor failure, preserve raw files and use quarantine. |
| Limited history/granularity | Start with narrow target; prefer baselines and disclose uncertainty. |
| Data does not support claimed feature | Do not infer field; leave unavailable until verified. |
| Forecast misuse | Model card, interval, horizon labels and non-operational disclaimer. |

| | |
|---|---|
| False anomaly interpretation | Signal-not-cause UI and human review workflow. |
| Carbon overclaim | Factor/boundary disclosure and estimate-only labeling. |
| LLM hallucination | Evidence retrieval, citation checker, refusal and audit logs. |
| Hackathon time | P0-first scope and static fallback demo. |

### 14.2 KPI framework

| Area | KPI | Status |
|---|---|---|
| Data | Ingestion success rate, validation pass rate, provenance completeness. | Target to be set after baseline measurement. |
| Forecast | MAE/RMSE/sMAPE, interval coverage/width by horizon versus baseline. | Measure; do not pre-claim. |
| Anomaly | Reviewed precision/recall where labels exist; review turnaround. | Future evaluation dependent on labels. |
| Product | Time to locate source and complete a scenario; task success rate. | Usability-study target. |
| Reliability | API availability, job latency, failure recovery time. | Operational target post-deployment. |
| Trust | Share of outputs with source/model/scenario references; user confidence feedback. | Governance metric. |

### 14.3 Judge differentiation

RE-PREDICT differentiates through epistemic discipline as much as features: it starts from a government-published reporting portal; preserves source lineage; compares forecasts against simple baselines; treats anomalies as signals rather than causal verdicts; makes scenario and carbon assumptions inspectable; and constrains AI narration to cited evidence. This is a credible path from hackathon prototype to decision-support product, not a dashboard with unsupported intelligence claims.

## 15. Demo, Judge Q and A and Presentation Plan

### 15.1 Demo storyline

| Moment | Narrative |
|---|---|
| 1. Context | Show that renewable reporting is valuable but analysis questions persist. |
| 2. Trust foundation | Open Source Studio: official report, retrieval date, checksum and lineage. |
| 3. Understand | Filter Insights dashboard and separate observed from derived data. |
| 4. Predict | Show a selected target, baseline comparison, ordered validation and interval. |
| 5. Detect | Open an anomaly; explain it as a deviation, not a confirmed cause. |
| 6. Simulate | Change declared capacity/availability assumptions and show equation/result. |
| 7. Explain | Ask AI Analyst a source-bounded question and reveal cited evidence. |
| 8. Close | Restate P0 evidence, guardrails, next validation steps and scalable vision. |

### 15.2 Judge questions and answers

| Question | Concise answer |
|---|---|
| Why CEA? | The cited portal is published by CEA under the Ministry of Power and labels Daily Renewable Generation Reports. RE-PREDICT preserves it as source data and does not claim to be CEA. |
| Is this an official CEA product? | No. It is an independent proposed decision-support layer with explicit source attribution. |
| How do you prevent hallucination? | The Analyst is retrieval-grounded, claim-typed, citation-checked and designed to refuse when evidence is missing. |

| | |
|---|---|
| How do you validate forecasts? | With ordered walk-forward folds, frozen data, leakage controls and comparison to seasonal-naive/persistence baselines. |
| Can an anomaly tell us why output changed? | No. It identifies deviation from an expected pattern; causal claims need independent evidence. |
| Are carbon savings verified? | No. Carbon Lens is an estimate framework only and discloses factor, boundary and counterfactual caveats. |
| Why not use deep learning now? | A simpler baseline/tree model is more defensible for MVP; LSTM/TFT is future work only if data and value justify it. |
| What makes this scalable? | Versioned contracts, lineage, APIs, model registry, monitoring and modular services make incremental scale possible. |

### 15.3 14 to 16 slide PPT structure

| Slide | Title | Content summary |
|---|---|---|
| 1 | Title | RE-PREDICT name, tagline and track. |
| 2 | The challenge | Renewable variability and the intelligence gap. |
| 3 | Why the source matters | CEA portal foundation and independent-layer positioning. |
| 4 | Our solution | Sense to Explain framework. |
| 5 | Who benefits | Users and representative decisions. |
| 6 | Product experience | Dashboard pages and trust UX. |
| 7 | Architecture | Source to API to analytics to dashboard diagram. |
| 8 | Data trust | Ingestion, validation and provenance. |
| 9 | Forecasting | Target, baselines, tree models and walk-forward validation. |
| 10 | Anomaly intelligence | Residual signal, explanation boundary and review. |
| 11 | Scenario and carbon | Equations, assumptions and caveats. |
| 12 | Responsible AI | Evidence-grounded Analyst and guardrails. |
| 13 | MVP and build plan | P0 scope, phases, 48-hour plan. |
| 14 | Validation and impact discipline | Acceptance tests, KPIs and limitations. |
| 15 | Demo journey | End-to-end story. |
| 16 | Closing | Vision, next step and final message. |

### 15.4 One-slide solution summary

RE-PREDICT is a proposed, source-traceable renewable-energy intelligence platform. It uses CEA-published renewable-generation reports as a data foundation, then adds validated normalization, trends, baseline-first forecasting, anomaly signals, transparent what-if scenarios and evidence-grounded explanations. It is an independent decision-support layer: observed values remain linked to their source, estimates show assumptions and uncertainty, and no output is presented as an official CEA determination.

### 15.5 Closing message

RE-PREDICT is not about claiming certainty from variable energy data. It is about making uncertainty visible, evidence traceable and planning conversations more rigorous. Sense the source. Understand the pattern. Predict with humility. Detect carefully. Simulate transparently. Explain with evidence.

## 16. References

1. Central Electricity Authority, Ministry of Power, Government of India. Renewable Generation Reports portal. https://gen-re.cea.gov.in/reports (accessed 02 October 2026). The portal identifies CEA and lists a Daily Renewable Generation Report with report names and dates.

2. Central Electricity Authority. Renewable Generation portal. https://gen-re.cea.gov.in/ (accessed 02 October 2026).

3. All equations, system designs, product modules, proposed metrics, scenarios, build plan and acceptance tests in this document are team-proposed unless explicitly described as source facts above.